Over the last couple of days I noticed that the labeling of the data sets seems to be mixed up. The data labeled as the collimated data has a pulse that arrives later than the focused data. I would expect this to be the other way around. This is noticeable in both the HDPE P polarised data, Fig. 1, and the Si wafer data sets, Fig. 2.

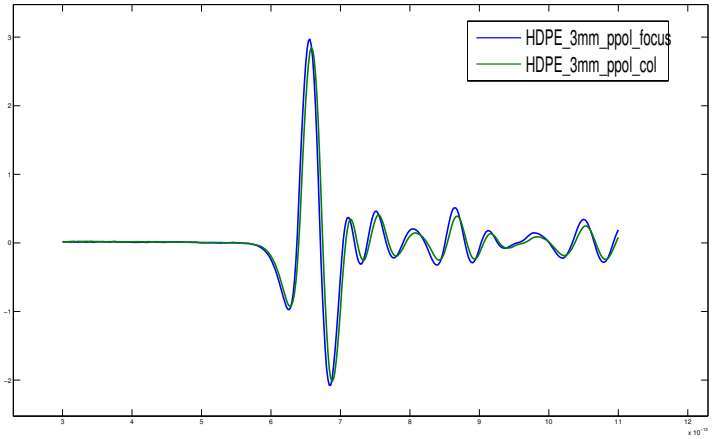

Figure 1: HDPE P polarisation time scans for the data labeled focused and collimated.

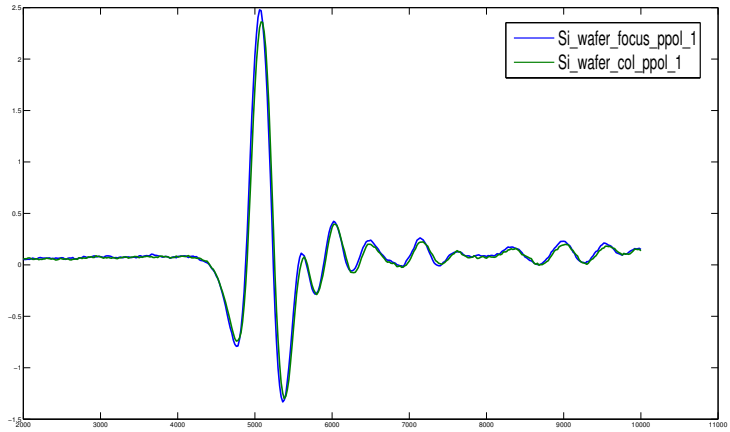

Figure 2: Si time scans for one pair of data labeled focused and collimated.

The HDPE data for the S polarisation does not actually show anytime shift, only attenuation, Fig. 3. This prevents any angle being extracted. In previous runs we had

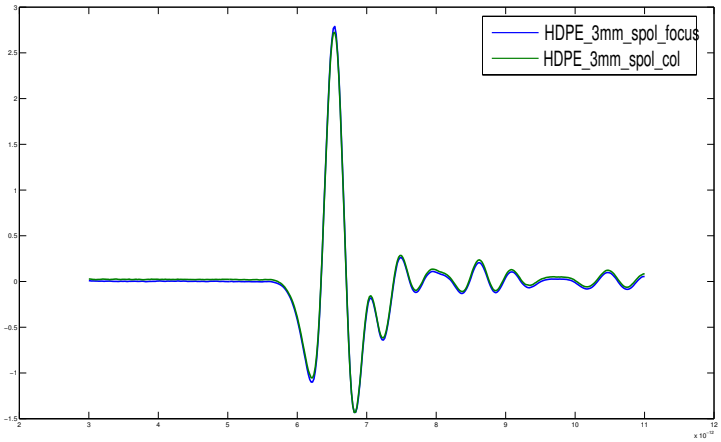

Figure 3: HDPE S polarisation time scans for the data labeled focused and collimated.

been using these data this way around as we assumed that the labels were correct.
I have now ran the extraction again using the data that arrives later as the focused data for all the data sets. The new angle extraction for HDPE P polarisation is shown in Fig. 4. As you can see the angle is now constant over the frequency range. The extraction

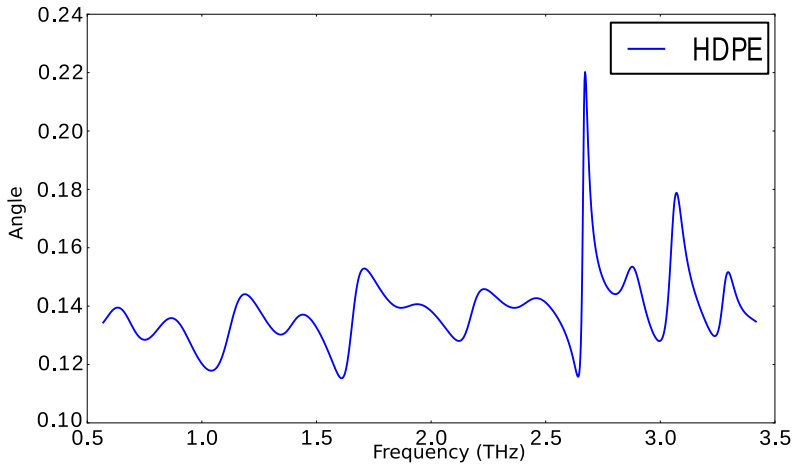

Figure 4: HDPE angle extraction.

however for the HDPE samples is tricky due to the absorptions from the sample which gives problems when doing the extrapolation of the phase in the algorithm. In short I am not sure how much we can trust the results from these sets.

The extraction I sent yesterday for the SI wafer is shown in Fig. 5. Again I am extracting with the focused data being the one where the pulse arrives last. As noted the angle

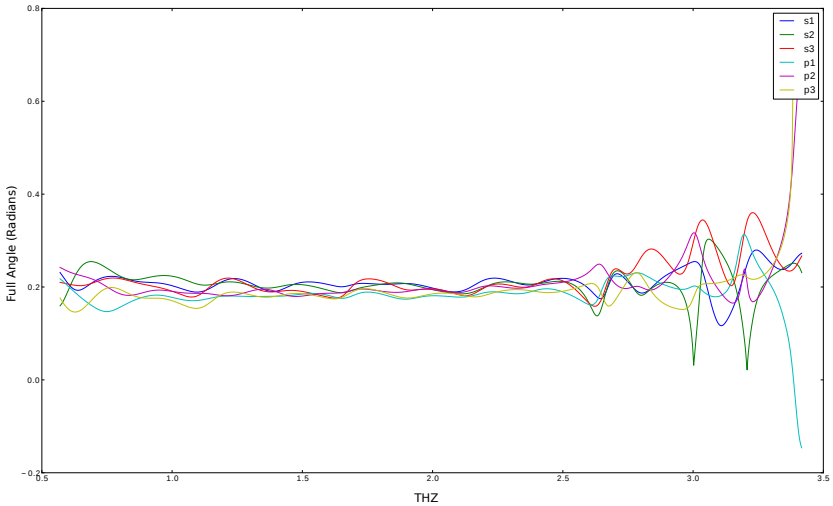

Figure 5: SI angle extraction.

does not change over this frequency range. This result is evident in the ratio of the transfer functions, T_r. Where,

$$T_r \simeq 1 - \frac{1}{2} i\alpha k\theta^2 \tag{1}$$

The phase of the transfer function for small angles is equal to its imaginary component.

$$\phi_r \simeq -\frac{1}{2}\alpha k\theta^2 \tag{2}$$

And thus the slope of the phase with respect to the frequency is

$$\frac{\Delta\phi_r}{\Delta f} = -\frac{1}{c}\pi\alpha\theta^2 \tag{3}$$

If the refractive index is constant then α is also constant and the slope only changes if θ changes. Which makes sense as you would expect a change in angle to look like dispersion. I have plotted the phase of T_r along with just the imaginary part to ensure this is valid. Figure. 6 shows the phase of the HDPE sample for P polarisation. Figure. 7 shows the phase of the HDPE sample for P polarisation. As can be seen in both cases the slope of the phase is constant over the region of interest and thus the change in angle (within our resolution) must be constant. Could it be that we are capping the maximum angle by the parabolics giving us a fixed angle? or could the change in angle over this range be to small for us to detect?

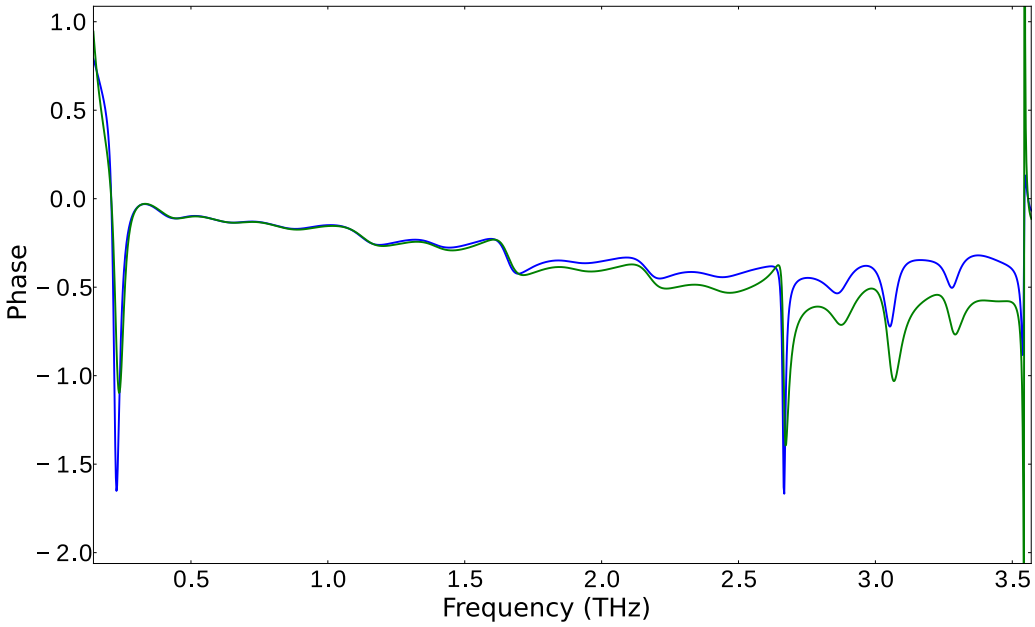

Figure 6: Phase of T_r for HDPE.

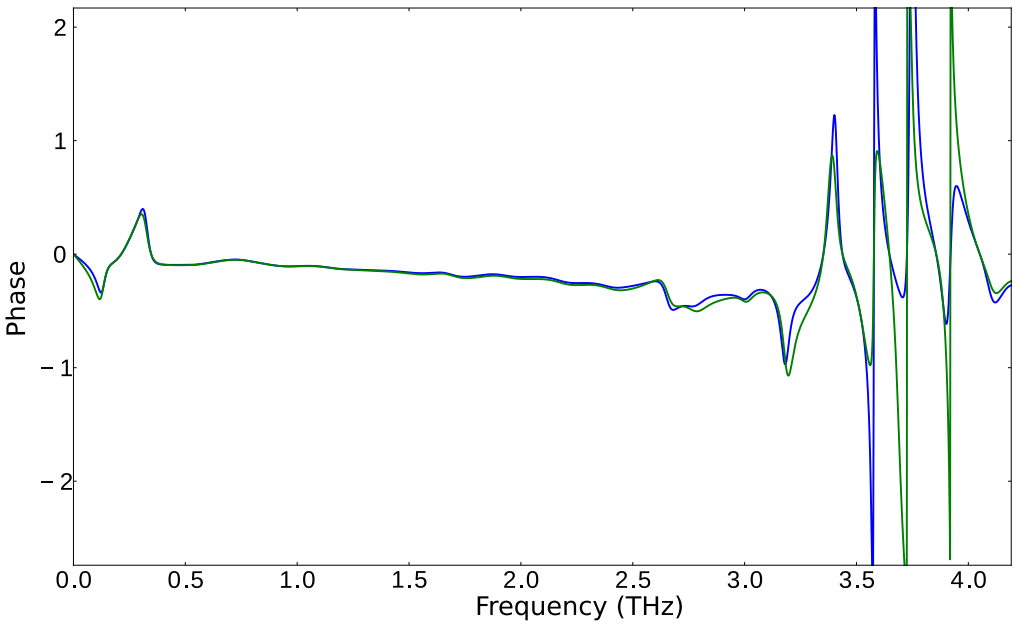

Figure 7: Phase of T_r for Si.